Exercises Week 5 (**with solutions**)

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1 Theoretical exercises

Exercise 5.1 Probabilistic PCA can be equivalently expressed as

$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \sigma^2\mathbf{I}) \tag{1}$$

(with latent variables $\mathbf{x}$) or

$$p(\mathbf{y}|\hat{\mathbf{x}}) = \mathcal{N}(\mathbf{y}|\hat{\mathbf{A}}\hat{\mathbf{x}} + \mathbf{b}, \sigma^2\mathbf{I}) \tag{2}$$

(with latent variables $\hat{\mathbf{x}} = \mathbf{R}\mathbf{x}$ and $\hat{\mathbf{A}} = \mathbf{A}\mathbf{R}^\intercal$, where $\mathbf{R}$ is a rotation matrix). The two latent representations $\mathbf{x}$ and $\hat{\mathbf{x}}$ are clearly different (for $\mathbf{R} \neq \mathbf{I}$). *Show that*

1. *Euclidean distances between points in these representations are identical, i.e.* $\|\mathbf{x}_i - \mathbf{x}_j\| = \|\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j\|$, *and*
2. *angles between points in these representations are identical, i.e.* $\angle(\mathbf{x}_i - \mathbf{x}_j, \mathbf{x}_k - \mathbf{x}_j) = \angle(\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j, \hat{\mathbf{x}}_k - \hat{\mathbf{x}}_j)$.

Solution 5.1 The first question is answered by a simple calculation

$$\begin{aligned}
\|\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j\| &= \|\mathbf{R}\mathbf{x}_i - \mathbf{R}\mathbf{x}_j\| && (3)\\
&= \|\mathbf{R}(\mathbf{x}_i - \mathbf{x}_j)\| && (4)\\
&= \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\intercal \mathbf{R}^\intercal \mathbf{R}(\mathbf{x}_i - \mathbf{x}_j)} && (5)\\
&= \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\intercal (\mathbf{x}_i - \mathbf{x}_j)} && (6)\\
&= \|\mathbf{x}_i - \mathbf{x}_j\| && (7)
\end{aligned}$$

Similarly for the second question

$$\begin{aligned}
\angle(\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j, \hat{\mathbf{x}}_k - \hat{\mathbf{x}}_j) &= \angle(\mathbf{R}(\mathbf{x}_i - \mathbf{x}_j), \mathbf{R}(\mathbf{x}_k - \mathbf{x}_j)) && (8)\\
&= \cos^{-1}\left(\frac{\langle \mathbf{R}(\mathbf{x}_i - \mathbf{x}_j), \mathbf{R}(\mathbf{x}_k - \mathbf{x}_j)\rangle}{\|\mathbf{R}(\mathbf{x}_i - \mathbf{x}_j)\| \; \|\mathbf{R}(\mathbf{x}_k - \mathbf{x}_j)\|}\right) && (9)\\
&= \cos^{-1}\left(\frac{(\mathbf{x}_i - \mathbf{x}_j)^\intercal \mathbf{R}^\intercal \mathbf{R}(\mathbf{x}_k - \mathbf{x}_j)}{\|(\mathbf{x}_i - \mathbf{x}_j)\| \; \|(\mathbf{x}_k - \mathbf{x}_j)\|}\right) && (10)\\
&= \cos^{-1}\left(\frac{(\mathbf{x}_i - \mathbf{x}_j)^\intercal (\mathbf{x}_k - \mathbf{x}_j)}{\|(\mathbf{x}_i - \mathbf{x}_j)\| \; \|(\mathbf{x}_k - \mathbf{x}_j)\|}\right) && (11)\\
&= \angle(\mathbf{x}_i - \mathbf{x}_j, \mathbf{x}_k - \mathbf{x}_j) && (12)
\end{aligned}$$

Exercise 5.2 Consider a generative model $\mathbf{y} = f(\mathbf{x})$, where $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$ and $\mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$. Let

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ f_3(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} 2x_1^2 + x_2^2 \\ x_1 \\ x_2 \end{pmatrix}. \tag{13}$$

1. *Derive the Jacobian matrix of f.*
2. *Show that the generative model spans an* immersed *manifold.*
3. *Show that the generative model spans an* embedded *manifold.*

Solution 5.2 For the first question, we derive the Jacobian one column at a time,

$$\mathbf{v}_1 = \frac{\partial f}{\partial x_1} = \begin{pmatrix} 4x_1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \frac{\partial f}{\partial x_2} = \begin{pmatrix} 2x_2 \\ 0 \\ 1 \end{pmatrix}. \tag{14}$$

The Jacobian is then

$$\mathbf{J} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{pmatrix} = \begin{pmatrix} 4x_1 & 2x_2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{15}$$

For the second question, we need to show that $\mathbf{J}$ is full rank for all $\mathbf{x}$. The columns of $\mathbf{J}$ are always non-zero, $\|\mathbf{v}_1\| > 0$ and $\|\mathbf{v}_2\| > 0$ for all $\mathbf{x}$. Next, we need to show that the columns of $\mathbf{J}$ are linearly independent, i.e. that $\mathbf{v}_1 \neq \alpha \mathbf{v}_2$ for $\alpha \neq 0$. This is trivially seen to be true, e.g. by looking at the 3rd row of the Jacobian.

For the third question, we note that the left-inverse of f (i.e. the inverse on its image) is trivially given by

$$f^{-1}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} y_2 \\ y_3 \end{pmatrix}. \tag{16}$$

The existence of the left-inverse ensures that f is injective, such that the manifold is embedded.

Exercise 5.3 Consider the curve $c : [0, 1] \to \mathbb{R}^2$ defined as

$$c(t) = \begin{pmatrix} 2t + 1 \\ -t^2 \end{pmatrix}. \tag{17}$$

1. *Derive an expression of the speed function* $t \mapsto \|\dot{c}_t\|$.
2. *Compute the Euclidean length of the curve. Hint:*

$$\int \sqrt{1 + t^2}\mathrm{d}t = \frac{1}{2}\left(\sqrt{1 + t^2}\ t + \sinh^{-1}(t)\right) + C, \tag{18}$$

for some constant C.

Solution 5.3 The derivative of c is

$$\dot{c}_t = \begin{pmatrix} 2 \\ -2t \end{pmatrix} \tag{19}$$

such that the speed function is

$$\|\dot{c}_t\| = 2\sqrt{1 + t^2}. \tag{20}$$

The curve length is then

$$\begin{aligned}
\text{length} &= \int_0^1 \|\dot{c}_t\|\mathrm{d}t && (21)\\
&= 2\int_0^1 \sqrt{1 + t^2}\mathrm{d}t && (22)\\
&= 2\left(\frac{1}{2}\left(\sqrt{1 + 1^2}\ 1 + \sinh^{-1}(1)\right) - \frac{1}{2}\left(\sqrt{1 + 0^2}\ 0 + \sinh^{-1}(0)\right)\right) && (23)\\
&= \sqrt{1 + 1^2} + \sinh^{-1}(1) && (24)\\
&= \sqrt{2} + \sinh^{-1}(1) && (25)\\
&\approx 2.2956 && (26)
\end{aligned}$$

Exercise 5.4 Let $x_1 \in (0, \pi)$ and $x_2 \in [0, 2\pi)$ be the latent coordinates of the manifold spanned by the function $f : (0, \pi) \times [0, 2\pi) \to \mathbb{R}^3$

$$f(x_1, x_2) = \begin{pmatrix} \sin(x_1)\cos(x_2) \\ \sin(x_1)\sin(x_2) \\ \cos(x_1) \end{pmatrix}. \tag{27}$$

This will span the unit sphere.

1. *Derive the Jacobian matrix of f.*
2. *Derive the metric associated with f according to Eq. 5.21 in the LMLG book.*
3. *Show that the metric is positive definite.*

Hint: recall that $\cos^2(x) + \sin^2(x) = 1$.

Solution 5.4 For the first question, we compute the Jacobian column by column,

$$\mathbf{v}_1 = \frac{\partial f}{\partial x_1} = \begin{pmatrix} \cos(x_1)\cos(x_2) \\ \cos(x_1)\sin(x_2) \\ -\sin(x_1) \end{pmatrix} \tag{28}$$

$$\mathbf{v}_2 = \frac{\partial f}{\partial x_2} = \begin{pmatrix} -\sin(x_1)\sin(x_2) \\ \sin(x_1)\cos(x_2) \\ 0 \end{pmatrix}, \tag{29}$$

such that the Jacobian is

$$\mathbf{J} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{pmatrix} = \begin{pmatrix} \cos(x_1)\cos(x_2) & -\sin(x_1)\sin(x_2) \\ \cos(x_1)\sin(x_2) & \sin(x_1)\cos(x_2) \\ -\sin(x_1) & 0 \end{pmatrix}. \tag{30}$$

For the second question, we compute the entries of the metric individually,

$$\mathbf{G} = \mathbf{J}^\intercal \mathbf{J} = \begin{pmatrix} \mathbf{v}_1^\intercal \mathbf{v}_1 & \mathbf{v}_1^\intercal \mathbf{v}_2 \\ \mathbf{v}_1^\intercal \mathbf{v}_2 & \mathbf{v}_2^\intercal \mathbf{v}_2 \end{pmatrix} \tag{31}$$

$$\mathbf{v}_1^\intercal \mathbf{v}_1 = \cos^2(x_1)\cos^2(x_2) + \cos^2(x_1)\sin^2(x_2) + \sin^2(x_1) \tag{32}$$

$$= \cos^2(x_1)\underbrace{\left(\cos^2(x_2) + \sin^2(x_2)\right)}_{=1} + \sin^2(x_1) \tag{33}$$

$$= \cos^2(x_1) + \sin^2(x_1) = 1. \tag{34}$$

$$\begin{aligned} \mathbf{v}_1^\intercal \mathbf{v}_2 = &-\cos(x_1)\cos(x_2)\sin(x_1)\sin(x_2) \\ &+ \cos(x_1)\sin(x_2)\sin(x_1)\cos(x_2) \end{aligned} \tag{35}$$

$$= 0. \tag{36}$$

$$\mathbf{v}_2^\intercal \mathbf{v}_2 = \sin^2(x_1)\sin^2(x_2) + \sin^2(x_1)\cos^2(x_2) \tag{37}$$
$$= \sin^2(x_1)\left(\sin^2(x_2) + \cos^2(x_2)\right) \tag{38}$$
$$= \sin^2(x_1). \tag{39}$$

Combined, the metric is, thus,

$$\mathbf{G} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2(x_1) \end{pmatrix}. \tag{40}$$

To answer the third question, we note that the eigenvalues of $\mathbf{G}$ are (because the matrix is diagonal)

$$\lambda_1 = 1, \qquad \lambda_2 = \sin^2(x_1). \tag{41}$$

Since both are positive for $x_1 \in (0, \pi)$, $\mathbf{G}$ is positive definite.

2 Programming exercises

The main programming task of this week is to get a sense of how to measure curve lengths. Having an intuition of this is key to understanding the geometry of latent variable models.

Exercise 5.5 Consider the curve $c : [0, 1] \to \mathbb{R}^2$ defined in Eq. 17.

1. *Write a computer program that evaluates the length of the curve c using Eq. 4.2 in the LMLG book.*
2. *If you have completed exercise 5.3:*
   (a) *Did the numerical and the analytical results agree?*
   (b) *Use the analytic expression for $\|\dot{c}_t\|$ to write a computer program that evaluates the length of the curve using Eq. 4.5 in the LMLG book. Note that you need to approximate the integral with a sum.*

Solution 5.5 An example `pytorch` implementation:

```python
import torch

def c(t):
    _t = t.reshape(-1, 1)
    return torch.concat((2*_t + 1, -_t**2), dim=1)

def speed(t):
    return 2 * torch.sqrt(1 + t**2)

def curve_length(c, T):
    ct = c(T)  # |T|x2
    delta = ct[1:] - ct[:-1]  # (|T|-1)x2
```

```
    retval = torch.sum(delta.reshape(T.numel()-1, -1)**2, dim=1).sqrt().sum()  # scalar
    return retval

def integrate_speed(T):
    s = speed(T)  # |T|
    dt = T[1] - T[0]  # assuming a linspace
    return s.sum() * dt

print('The curve length is {}'.format(curve_length(c, torch.linspace(0, 1, 100))))
print('The integrated speed  is {}'.format(integrate_speed(torch.linspace(0, 1, 100))))
```

Exercise 5.6 Consider the Bernoulli VAE that you worked on in Week 1. Train this with a two-dimensional latent space (for ease of plotting).

1. *Write a computer program that evaluates the length of any latent second-order polynomial curve c using Eq. 4.2 in the LMLG book. It is recommended that you write the code to support any callable curve c.*

Solution 5.6 See `dggm_ex56.py`.